

Pedro G. S. Fernandes

Curriculum Vitae

✉ fernandes@thphys.uni-heidelberg.de
🔗 inspire-hep, Google Scholar
🌐 Personal Website
🐙 GitHub

EMPLOYMENT

OCT 2024 – PRESENT	Postdoctoral Researcher Institut für Theoretische Physik <i>Heidelberg University, Germany</i>
OCT 2023 – OCT 2024	Postdoctoral Researcher CP3-Origins, Quantum Gravity Group <i>University of Southern Denmark, Denmark</i>
OCT 2022 – SET 2023	Postdoctoral Researcher Particle Cosmology Group <i>University of Nottingham, UK</i>
NOV 2019 – MAR 2023	PhD Researcher LINK TO DISSERTATION Supervisors: David Mulryne and Timothy Clifton <i>Queen Mary University of London</i>

EDUCATION

2017 – 2019	Master of Science in Physics LINK TO DISSERTATION Supervisors: Carlos Herdeiro and Eugen Radu <i>Instituto Superior Técnico, University of Lisbon</i>
2014 – 2017	Bachelor of Physics <i>Instituto Superior Técnico, University of Lisbon</i>

TEACHING EXPERIENCE

2024-2026	Quantum Field Theory <i>University of Heidelberg</i>
2020-2023	Statistical Physics; Thermodynamics; Quantum Mechanics; Astrophysics <i>Queen Mary University of London</i>
2019	Thermodynamics and Structure of Matter <u>Excellence in teaching award</u> <i>Instituto Superior Técnico, University of Lisbon</i>

ACADEMIC SERVICE

- Invited to referee for *Physical Review Letters*, *Physical Review D*, *Classical and Quantum Gravity*, *Physics Letters B*, *JCAP*, *European Physical Journal C*, *General Relativity and Gravitation*, and others.
- Member of the LISA consortium.
- Member of the Portuguese Society for Relativity and Gravitation.

PUBLICATIONS

Full list of publications available on my [inspire-hep](#) and [Google Scholar](#) profiles. **Bibliometric metrics:** 29 papers, 1469 citations, h-index=15 (data from inspire-hep). [Orcid](#): 0000-0002-8176-7208. [Scopus Author ID](#): 57209776012.

AWARDS AND FELLOWSHIPS

OCT 2024	DFG Fellowship EXC 2181/1 - 390900948 <i>University of Heidelberg</i>
OCT 2023	Villum Fonden Fellowship NO. 29405 <i>University of Southern Denmark</i>
OCT 2022	Leverhulme Trust Fellowship <i>University of Nottingham</i>
NOV 2019	Royal Society PhD Grant RGF/EA/180022 <i>Queen Mary University of London</i>
2019	FCT-CERN Research Grant CERN/FIS-PAR/0027/2019 <i>Collaborator, FCT-CERN</i>
2019	Excellence in Teaching Award <i>Instituto Superior Técnico, U. Lisbon</i>
FEB 2019 – JUL 2019	Teaching Fellowship <i>Instituto Superior Técnico, U. Lisbon</i>
JUL 2018 – JAN 2019	Research Fellowship UID/CTM/04540/2013 CEFEMA, RD 0472 Department of Physics <i>Instituto Superior Técnico, U. Lisbon</i>
JUN 2017 – JUN 2018	Research Fellowship SCIENTIFIC INITIATION GRANT Department of Physics <i>Instituto Superior Técnico, U. Lisbon</i>

COMPUTER SKILLS AND CODE

C, C++, Python, Julia, Mathematica
Author of *SpinningBlackHoles*, a publicly available code to obtain stationary, axially-symmetric black hole solutions in modified gravity. See also my [GitHub](#).

MENTORING

Heidelberg University

- 2024-2025 Lidia Marino (MSc), co-advisor, now PhD at Heidelberg University
- 2024-2026 Fabian Willaschek (MSc), co-advisor
- 2025 Nadine Nussbaumer (PhD), informal co-advisor
- 2025 Jingqian Gou (PhD), informal co-advisor
- 2025-2026 Verena Köder (MSc), co-advisor

SELECTED CONFERENCES AND TALKS

- XVIII Black Holes Workshop, Instituto Superior Técnico, Lisbon, speaker, Dec 2025
- Crossroads in Strong Gravity, University of Catania, Italy, session chair, Sep 2025
- University of Concepción, Chile, invited seminar, May 2025
- CENTRA, Instituto Superior Técnico, Lisbon, invited seminar, Feb 2025
- Gravity Shape Pisa 2024, invited keynote speaker, Oct 2024
- 50 Years of Horndeski Gravity, Perimeter Institute for Theoretical Physics, speaker, Jul 2024
- Université Paris-Saclay, IJCLab, invited seminar, Feb 2024
- AstroParticle Symposium 2023, Institut Pascal, Paris, invited keynote speaker, Nov 2023
- Nottingham Centre of Gravity Meeting, University of Nottingham, invited seminar, Apr 2023
- Imperial College London, invited seminar, Feb 2023
- XV Black Holes Workshop, ISCTE, Lisbon, keynote speaker, Dec 2022
- Gravity @ Prague, Charles University Prague, Attendant, Set 2022
- Center for Gravitation and Cosmology, Yangzhou University, invited webinar, Aug 2022
- London Cosmology Discussion Meeting (LCDM), invited speaker, Dec 2021
- 50th BUSSTEPP School, Queen Mary University of London, speaker, Jan 2021 (*award for best session talk*)
- XIII Black Holes Workshop, Instituto Superior Técnico, speaker, Dec 2020
- Quantum Gravity group, University of Groningen, invited webinar, May 2020
- Gravitational Geometry and Dynamics group, University of Aveiro, invited webinar, May 2020
- COSMONATA, Faculty of Sciences University of Lisbon, invited seminar, Dec 2019
- 4th CENTRA Meeting, Faculty of Sciences University of Lisbon, speaker, Mar 2019
- XI Black Holes Workshop, Instituto Superior Técnico, Lisbon, Attendant, Dec 2018

LIST OF PUBLICATIONS

- [1] Pedro G. S. Fernandes. Leading effective field theory corrections to the Kerr metric at all spins, 12 2025. [arXiv:2512.02338](#).
- [2] Pedro G. S. Fernandes, Jingqian Gou, Lavinia Heisenberg, and Nadine Nussbaumer. Inflation, black holes with primary hair, and regular planar black holes from an infinite tower of regularized Lovelock-Proca corrections, 11 2025. [arXiv:2511.22798](#).
- [3] Kyriakos Destounis and Pedro G. S. Fernandes. Chaos and Carter: Extreme-mass-ratio systems of relativistic rotating black holes in astrophysical environments, 8 2025. [arXiv:2508.20191](#).
- [4] Astrid Eichhorn and Pedro G. S. Fernandes. Regular black holes without mass-inflation instability and gravastars from modified gravity, 8 2025. [arXiv:2508.00686](#). Submitted to PRL.
- [5] Pedro G. S. Fernandes and Vitor Cardoso. Spinning Black Holes in Astrophysical Environments. *Phys. Rev. Lett.*, 135(21):211403, 2025. [arXiv:2507.04389](#).

- [6] Farid Thaalba, Pedro G. S. Fernandes, and Thomas P. Sotiriou. Supermassive black hole scalarization and effective field theory, 6 2025. [arXiv:2506.21434](#).
- [7] Raúl Carballo-Rubio, Héloïse Delaporte, Astrid Eichhorn, and Pedro G. S. Fernandes. Nonminimal light-curvature couplings and black-hole imaging. *Phys. Rev. D*, 112(10):103016, 2025. [arXiv:2505.21431](#).
- [8] Pedro G. S. Fernandes and Vitor Cardoso. Dark matter as a vector field: an action principle for the Einstein cluster, 5 2025. [arXiv:2505.00563](#).
- [9] Christos Charmousis, Pedro G. S. Fernandes, and Mokhtar Hassaine. Proca theory of four-dimensional regularized Gauss-Bonnet gravity and black holes with primary hair. *Phys. Rev. D*, 111(12):124008, 2025. [arXiv:2504.13084](#).
- [10] Pedro G. S. Fernandes. Regular BTZ black holes from an infinite tower of corrections. *Phys. Lett. B*, 868:139772, 2025. [arXiv:2504.08565](#).
- [11] Pedro G. S. Fernandes. Singularity resolution and inflation from an infinite tower of regularized curvature corrections. *Phys. Rev. D*, 112(8):084028, 2025. [arXiv:2504.07692](#).
- [12] Kalin V. Staykov, Daniela D. Doneva, Pedro G. S. Fernandes, and Stoytcho S. Yazadjiev. Rotating scalarized black holes: The role of the coupling. *Phys. Rev. D*, 112(2):024058, 2025. [arXiv:2503.15206](#).
- [13] Astrid Eichhorn and Pedro G. S. Fernandes. Purely metric Horndeski theories and spontaneous curvaturization of black holes. *Phys. Rev. D*, 111(10):104007, 2025. [arXiv:2502.14717](#).
- [14] Giulia Ventagli, Pedro G. S. Fernandes, Andrea Maselli, Antonio Padilla, and Thomas P. Sotiriou. Neutron stars and the cosmological constant problem. *Phys. Rev. D*, 111(2):024001, 2025. [arXiv:2404.19012](#).
- [15] Pedro G. S. Fernandes, Clare Burrage, Astrid Eichhorn, and Thomas P. Sotiriou. Shadows and properties of spin-induced scalarized black holes with and without a Ricci coupling. *Phys. Rev. D*, 109(10):104033, 2024. [arXiv:2403.14596](#).
- [16] Astrid Eichhorn, Pedro G. S. Fernandes, Aaron Held, and Hector O. Silva. Breaking black-hole uniqueness at supermassive scales. *Class. Quant. Grav.*, 42(10):105006, 2025. [arXiv:2312.11430](#).
- [17] Clare Burrage, Pedro G. S. Fernandes, Richard Brito, and Vitor Cardoso. Spinning black holes with axion hair. *Class. Quant. Grav.*, 40(20):205021, 2023. [arXiv:2306.03662](#).
- [18] Pedro G. S. Fernandes. Rotating black holes in semiclassical gravity. *Phys. Rev. D*, 108(6):L061502, 2023. [arXiv:2305.10382](#).
- [19] Pedro G. S. Fernandes and David J. Mulryne. A new approach and code for spinning black holes in modified gravity. *Class. Quant. Grav.*, 40(16):165001, 2023. [arXiv:2212.07293](#).
- [20] Pedro G. S. Fernandes, David J. Mulryne, and Jorge F. M. Delgado. Exploring the Small Mass Limit of Stationary Black Holes in Theories with Gauss-Bonnet Terms. *Class. Quant. Grav.*, 39:235015, 2022. [arXiv:2207.10692](#).
- [21] Pedro G. S. Fernandes, Pedro Carrilho, Timothy Clifton, and David J. Mulryne. The 4D Einstein–Gauss–Bonnet theory of gravity: a review. *Class. Quant. Grav.*, 39(6):063001, 2022. [arXiv:2202.13908](#).
- [22] Pedro G. S. Fernandes, Pedro Carrilho, Timothy Clifton, and David J. Mulryne. Black holes in the scalar-tensor formulation of 4D Einstein-Gauss-Bonnet gravity: Uniqueness of solutions, and a new candidate for dark matter. *Phys. Rev. D*, 104(4):044029, 2021. [arXiv:2107.00046](#).
- [23] Pedro G. S. Fernandes. Gravity with a generalized conformal scalar field: theory and solutions. *Phys. Rev. D*, 103(10):104065, 2021. [arXiv:2105.04687](#).
- [24] Timothy Clifton, Pedro Carrilho, Pedro G. S. Fernandes, and David J. Mulryne. Observational Constraints on the Regularized 4D Einstein-Gauss-Bonnet Theory of Gravity. *Phys. Rev. D*, 102(8):084005, 2020. [arXiv:2006.15017](#).
- [25] Pedro G. S. Fernandes, Pedro Carrilho, Timothy Clifton, and David J. Mulryne. Derivation of Regularized Field Equations for the Einstein-Gauss-Bonnet Theory in Four Dimensions. *Phys. Rev. D*, 102(2):024025, 2020. [arXiv:2004.08362](#).
- [26] Pedro G. S. Fernandes. Charged black holes in AdS spaces in 4D Einstein Gauss-Bonnet gravity. *Phys. Lett. B*, 805:135468, 2020. [arXiv:2003.05491](#).
- [27] Pedro G. S. Fernandes. Einstein–Maxwell-scalar black holes with massive and self-interacting scalar hair. *Phys. Dark Univ.*, 30:100716, 2020. [arXiv:2003.01045](#).
- [28] Pedro G. S. Fernandes, Carlos A. R. Herdeiro, Alexandre M. Pombo, Eugen Radu, and Nicolas Sanchis-Gual. Charged black holes with axionic-type couplings: Classes of solutions and dynamical scalarization. *Phys. Rev. D*, 100(8):084045, 2019. [arXiv:1908.00037](#).
- [29] Pedro G. S. Fernandes, Carlos A. R. Herdeiro, Alexandre M. Pombo, Eugen Radu, and Nicolas Sanchis-Gual. Spontaneous Scalarisation of Charged Black Holes: Coupling Dependence and Dynamical Features. *Class. Quant. Grav.*, 36(13):134002, 2019. [arXiv:1902.05079](#).